

Experiment - 5.1.1. Leap Year Checker

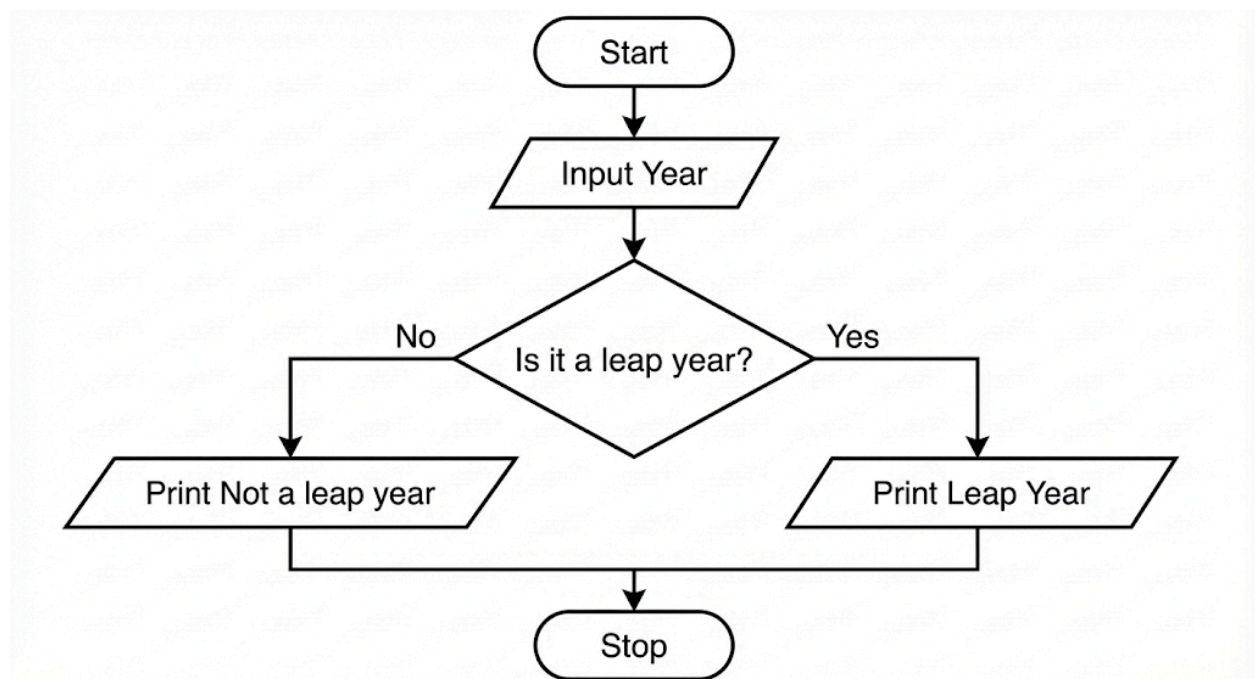
1. Aim

To design and implement a Python program that prompts the user to enter a year and determines whether the given year is a leap year or not, printing the appropriate message.

2. Pseudocode

1. **START**
2. **READ** the input value from the user and convert it to an integer.
3. **STORE** this value in the variable `y`.
4. **CHECK** if the year `y` is divisible by 400 ($y \% 400 == 0$).
 - **OR** check if the year `y` is divisible by 4 AND not divisible by 100 ($y \% 4 == 0$ and $y \% 100 != 0$).
5. **IF** the condition in step 4 is True, **PRINT** "Leap year".
6. **ELSE, PRINT** "Not a leap year".
7. **END**

3. Flowchart



4. Python Program

```
y = int(input())
```

```
if (y % 400 == 0) or (y % 4 == 0 and y % 100 != 0):  
    print("Leap year")  
else:  
    print("Not a leap year")
```

5. Experiment Screenshot

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5.1.1. Leap Year Checker 0.003 s

Write a Python program that prompts the user to enter a year. The program should determine if the year is a leap year or not and print the appropriate message.

Input Format:

- A single line contains an integer representing the year.

Output Format:

- Print "Leap year" if it is a leap year. Otherwise, print "Not a leap year".

Sample Test Cases +

```
1 y = int(input())  
2  
3 if (y % 400 == 0) or (y % 4 == 0 and y % 100 != 0):  
4     print("Leap year")  
5 else:  
6     print("Not a leap year")  
7  
8
```

Average time: **0.003 s** Maximum time: **0.004 s**

3.00 ms 4.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 4 ms Debug Test cases

Expected output
2024
Leap year

Actual output
2024
Leap year

Terminal Test cases

< Prev Reset Submit Next >

Experiment - 5.1.2. Student Grade Based on Aggregate

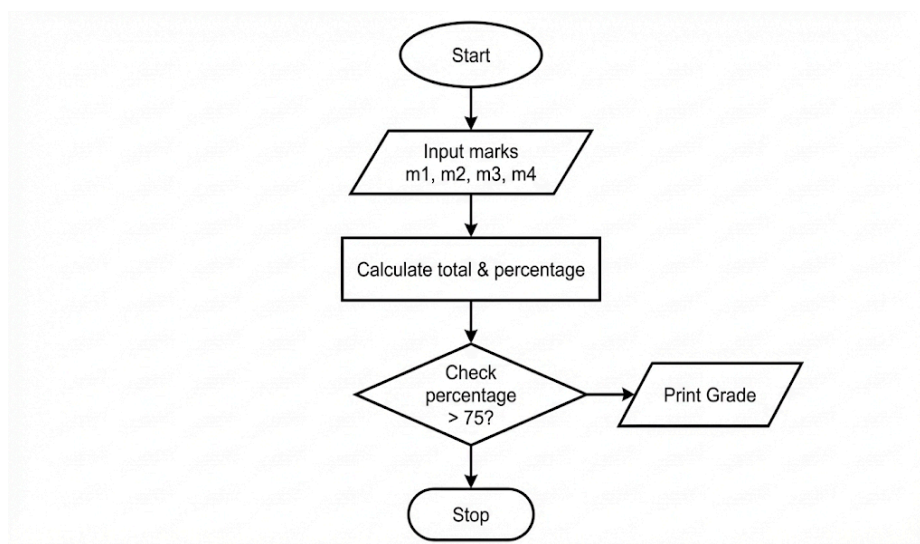
1. Aim

To design and implement a Python program that calculates a student's total marks, aggregate percentage, and final grade based on the marks obtained in four subjects. The program determines the grade using a specific set of percentage-based conditions.

2. Pseudocode

1. **START**
2. **READ** four space-separated integers from the user and **STORE** them in variables m1, m2, m3, and m4.
3. **CALCULATE** the sum of the four subjects and **STORE** it in the variable total (total = m1 + m2 + m3 + m4).
4. **CALCULATE** the aggregate percentage by dividing the total by 4 and **STORE** it in the variable percentage.
5. **CHECK** the value of percentage to determine the grade:
 - **IF** percentage > 75, **STORE** "Distinction" in grade.
 - **ELSE IF** percentage >= 60, **STORE** "First Division" in grade.
 - **ELSE IF** percentage >= 50, **STORE** "Second Division" in grade.
 - **ELSE IF** percentage >= 40, **STORE** "Third Division" in grade.
 - **ELSE**, **STORE** "Fail" in grade.
6. **PRINT** the total marks.
7. **PRINT** the percentage formatted to two decimal places.
8. **PRINT** the grade.
9. **END**

3. Flowchart



4. Python Program

```
# Read four subject marks
m1, m2, m3, m4 = map(int, input().split())
```

```
# Calculate total and percentage
total = m1 + m2 + m3 + m4
percentage = total / 4
```

```
# Determine the grade
if percentage > 75:
    grade = "Distinction"
elif percentage >= 60:
    grade = "First Division"
elif percentage >= 50:
    grade = "Second Division"
elif percentage >= 40:
    grade = "Third Division"
else:
    grade = "Fail"
```

```
# Output the results
print(total)
print(f"{percentage:.2f}")
print(grade)
```

5. Experiment Screenshot

The screenshot displays the CODETANTRA IDE interface. On the left, a sidebar contains the file explorer and a list of files. The main editor area shows a Python program for calculating student grades. The program reads four subject marks, calculates the total and percentage, and determines the grade based on the percentage. The IDE interface includes a file explorer, a code editor, and a terminal/output window.

Program Description: Write a program to calculate the total marks, aggregate percentage, and grade of a student based on marks in four subjects. The grade is determined as follows:

- Aggregate > 75%: Distinction
- Aggregate >= 60% and < 75%: First Division
- Aggregate >= 50% and < 60%: Second Division
- Aggregate >= 40% and < 50%: Third Division
- Aggregate < 40%: Fail

Input Format:

- Four space-separated integers representing the marks in four subjects.

Output Format:

- The first line should print the total marks.
- The second line should print the aggregate percentage with two decimal places.
- The third line should print the grade.

Constraints:

- 0 <= marks in each subject <= 100

Sample Test Cases

Test Case 1:

Expected output	Actual output
341	341
85.25	85.25
Distinction	Distinction